Problem 1

Game 2

$$\int_{-1}^{0} 5(x^6 + x)^4 \, \mathrm{d}x$$

The UK University Integration Bee

2024/25

Chainlink Round

Round 3

Problem 2

Game 2

$$\int e^{e^{e^x}} e^{e^x} e^{e^x} e^x \, \mathrm{d}x$$

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Chainlink Round

Round 3

Problem 3

Game 2

$$\int_{0}^{\frac{\pi}{2}} \sec(x) \csc(x) \sin(\tan(x)) \, \mathrm{d}x$$

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Round 3

Problem 4

Game 2

$$\int_{0}^{\infty} \ln(1 + e^{-x}) \cos(x) \, \mathrm{d}x$$

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Round 3

Problem 5

Game 2

$$\int_0^{\frac{\pi}{2}} \frac{\sin(x)}{\cos(x)+\tan(x)}\,\mathrm{d}x$$

Problem 6

Game 2

$$\int_0^{\infty} e^{-a^2x^2-\frac{b^2}{x^2}}\,\mathrm{d}x$$

$$\text{for } a,b>0$$

Problem 7

Game 2

$$\int_0^{\frac{\pi}{2}} \frac{\cos^3(x)\sin(x)}{\ln(\sin(x))}\,\mathrm{d}x$$

Problem 8

Game 2

$$\lim_{n\to\infty}\int_0^1 2025\sin(\{nx^{2025}\})\cos(\{nx^{2025}\})\,\mathrm{d}x$$